Emir Ahmed

New Haven, CT | (203) 508-5391 | emir.ahmed@yale.edu | Portfolio | GitHub | LinkedIn | Music

Education

Yale University | *B.S. in Electrical Engineering and Computer Science* — Expected 2028

American College of Sofia | *Hardware & Software Technologies profile; Floyd Black Award* — 2024

Experience

Yale Undergraduate Robotics, Mars Rover | *Communications & Control Subteam Lead* — Sept 2026 – Present

- Lead wireless communications, telemetry and teleoperation for Yale's University Rover Challenge 2027 rover.
- Designed the rover-to-base radio architecture (2.4 GHz Ubiquiti main link, 915 MHz backup) to fit URC band rules and FCC limits; specified and procured the full RF system for about $590.
- Built a ROS 2 bridge for commands and telemetry over a low-bandwidth serial radio (packet framing, CRC checks, rate limiting); set up the rover and base-station Raspberry Pi 5s on ROS 2 Jazzy.

Yale Interactive Machines Group | *Research Assistant & Developer, Dr. Marynel Vázquez*

- **Shutter:** own the full migration of the codebase of a robot photographer being deployed at the Yale Peabody Museum.
- **Youwrite.ai:** building a writing platform for researchers: a flow graph of ideas kept in sync with a text editor, with an LLM that acts as a supervisor and asks questions when the argument is unclear instead of writing for the author.

Building Interactive Machines, Yale Computer Science | *Undergraduate Learning Assistant*

- Help write assignments, hold office hours, and set up the machines and robots students use in class.

Yale School of Management | *Research Assistant*

- Cleaned a large database of historical London Stock Exchange data against scanned source documents; built a desktop tool that opens the matching PDF page from copied Excel cells, and fine-tuned an LLM on Vertex AI to automate part of the manual checking.

Yale Department of Near Eastern Languages & Civilizations | *Web Developer*

- Building the Turkish program's website and a student site with a resource library, flashcards and AI-generated lesson videos.

Projects

Manuscript | *creator and sole developer* / *manuscript.video* · *Example videos* — 2026

- Web app that turns a music score and a recording into a synced score-animation video, with every note lighting up as it sounds.
- Built a fully client-side rendering pipeline: Verovio (WASM) engraving, a beat-domain timing core with user-placed sync anchors, and MP4 export up to 4K via WebCodecs, with no server rendering.
- Implemented an automatic audio-to-score aligner, GPU-warped previews, and a parity test harness (Vitest) that keeps preview and export frame-identical; shipped end to end on Next.js, TypeScript and Firebase.

Gravity Pulls | *weighted ball that composes a piano piece from how it rolls* / *Demo* · *Code* · *Write-up* — Nov 2026

- Sealed sphere with an nRF52840 IMU reports its floor-contact tile over BLE; a Raspberry Pi composes a piano piece per roll (tiles to melody, spin to mode, wall hits to meter) with a rhythm state machine, Viterbi voice leading and FluidSynth.
- Custom wall-hit sensor: two piezos on an ESP32-S3, each hit confirmed against the ball's accelerometer within 100 ms.

iSpy | *Yale x SOM Hackathon 2026* / *Code* — Apr 2026

- Co-built a voice-driven AI vision assistant for blind users: a FastAPI server runs a Google ADK agent that answers spoken questions about a live camera feed with Gemini and replies through ElevenLabs.
- Fully hands-free: offline Vosk wake word, a "watch for" mode that alerts when a requested object appears, and session memory for follow-ups.

PosturePoke | *5th place, YHack 2026 Hardware Track* / *Demo* · *Code* — Mar 2026

- Co-built a wearable posture monitor that streams IMU data at 10 Hz from an Arduino to a web app over the Web Serial API and escalates from beeps to an AI voice coach to a servo-powered poke when the user slouches.
- Generated custom AI coaches from a text description (Gemini scripts, ElevenLabs voices), with social features on Firebase.

Pec Synth | *wearable instrument played by flexing the pecs* / *Demo* · *Code* · *Write-up* — 2026

- Two nRF52840 IMU boards stream motion to a Python host on a Raspberry Pi that detects flexes and plays melodies in Turkish maqams. Tuned detection to 53/54 on a labeled recording and cut flex-to-sound latency from about 300 ms to under 100 ms.

Skills

Languages: TypeScript, JavaScript, Python, C, C++, C#, SQL, Racket
Robotics & Embedded: ROS 2, MoveIt, Arduino, Raspberry Pi, ESP32, nRF52840
Web & Desktop: React, Next.js, Tailwind CSS, Firebase, Supabase, WPF, PySide6
AI & Data: OpenAI API, Gemini, ElevenLabs, LLM fine-tuning (Vertex AI), NumPy, pandas, scikit-learn